

Benchmarking ML Approaches over Different Dataset

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Agenda

1. Dataset & Task Overview
2. Model Architectures
3. Iris Dataset: Characteristics & Architectures & Results
4. Wine Dataset: Characteristics & Architectures & Results
5. Adult Dataset: Characteristics & Architectures & Results
6. Comparative Analysis & Interpretation
7. Final Conclusion



Dataset & Task Overview

- ❖ **Datasets:** Iris (simple), Wine (medium), Adult (complex)
- ❖ **Models Evaluated:**
 - Gaussian Naive Bayes
 - Decision Tree Classifier
 - Multi-Layer Perceptron (MLP)
 - LLM with Few-Shot / In-Context Learning
- ❖ **Objective:**
Evaluate performance, architecture behavior, generalization ability, and limitations across dataset complexities.

Variables Table						
Variable Name	Role	Type	Description	Units	Missing Values	
class	Target	Categorical			no	
Alcohol	Feature	Continuous			no	
Malicacid	Feature	Continuous			no	
Ash	Feature	Continuous			no	
Alcalinity_of_ash	Feature	Continuous			no	
Magnesium	Feature	Integer			no	
Total_Phenols	Feature	Continuous			no	
Flavanoids	Feature	Continuous			no	
Nonflavanoid_phenols	Feature	Continuous			no	
Proanthocyanins	Feature	Continuous			no	
Color_intensity	Feature	Continuous			no	
Hue	Feature	Continuous			no	
00280_00315_of_diluted_wines	Feature	Continuous			no	
Proline	Feature	Integer			no	

Variables Table						
Variable Name	Role	Type	Demographic	Description	Units	Missing Values
age	Feature	Integer	Age	N/A		no
workclass	Feature	Categorical	Income	Private, Self-emp-not-inc, Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.		yes
fnlwgt	Feature	Integer				no
education	Feature	Categorical	Education Level	Bachelors, Some-college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th, Preschool.		no
education-num	Feature	Integer	Education Level			no
marital-status	Feature	Categorical	Other	Married-div-spouse, Divorced, Never-married, Separated, Widowed, Married-spouse-absent, Married-AF-spouse.		no
occupation	Feature	Categorical	Other	Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-serv, Armed-Forces.		yes
relationship	Feature	Categorical	Other	Wife, Own-child, Husband, Not-in-family, Other-relative, Unmarried.		no
race	Feature	Categorical	Race	White, Asian-Pac-Islander, Amer-Indian-Eskimo, Other, Black.		no
sex	Feature	Binary	Sex	Female, Male.		no
capital-gain	Feature	Integer				no
capital-loss	Feature	Integer				no
hours-per-week	Feature	Integer				no
native-country	Feature	Categorical	Other	United-States, Cambodia, England, Puerto-Rico, Canada, Germany, Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland, Thailand, Yugoslavia, El-Salvador, Trinidad&Tobago, Peru, Hong, Holand-Netherlands.		yes
income	Target	Binary	Income	>50K, <=50K.		no

Variables Table						
Variable Name	Role	Type	Description	Units	Missing Values	
sepal length	Feature	Continuous		cm	no	
sepal width	Feature	Continuous		cm	no	
petal length	Feature	Continuous		cm	no	
petal width	Feature	Continuous		cm	no	
class	Target	Categorical	class of iris plant: Iris Setosa, Iris Versicolour, or Iris Virginica		no	

❖ Gaussian Naive Bayes

- Probabilistic model assuming **feature-wise conditional independence**
- Each feature $\sim$ Gaussian distribution
- Parameters estimated analytically $\rightarrow$ fastest training
- Works best when features are weakly correlated

❖ Decision Tree (CART)

- Recursive binary splits using **Gini impurity**
- Learns hierarchical rules
- Built-in feature selection
- Easily interpretable (tree structure)
- Sensitive to depth $\rightarrow$ pre-pruning used

❖ MLP Classifier

- Feed-forward neural network
- Hidden layer learns **non-linear decision boundaries**
- Activation: **ReLU**
- Optimizers used: Adam/LBFGS
- Regularized using **L2 penalty (alpha)**
- Can overfit without tuning

❖ LLM (In-Context Learning)

- No training; only reasoning from examples
- Works as pattern recognizer
- Not optimized for numeric/tabular patterns
- Performance depends on how well patterns are represented in prompt

Iris Dataset: Characteristics & Architectures



❖ Dataset Characteristics

- 150 samples, 4 numeric features
- 3-class classification (Setosa, Versicolor, Virginica)
- Clear linear separability

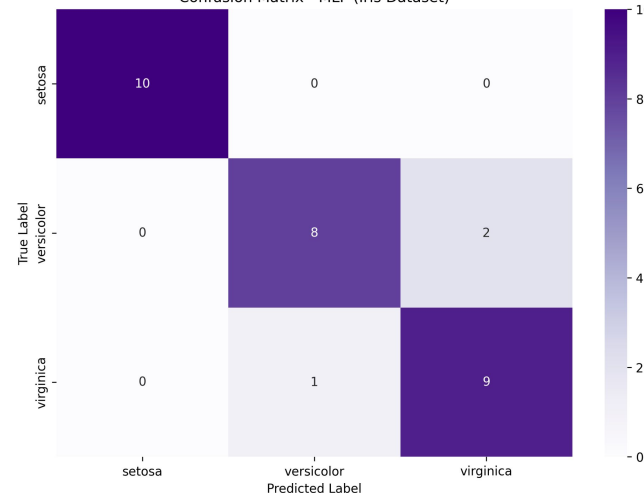
❖ Model Architectures Used

- **Naive Bayes:** Gaussian assumption fits well
- **Decision Tree:** Depth=5 → perfectly captures petal thresholds
- **MLP:** 1 hidden layer (100 neurons), ReLU
- **LLM-ICL:** 3–5 demonstration examples achieve high pattern matching

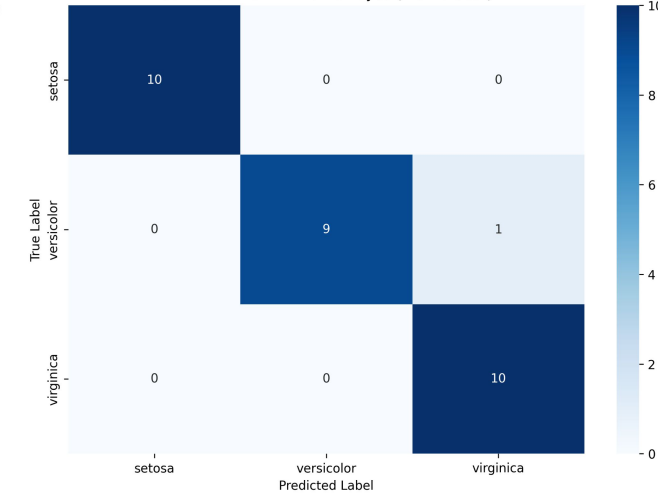
❖ Why Models Perform Well

- Features (especially petal length/width) form natural clusters
- Low noise, well-separated boundaries → ideal for all algorithms

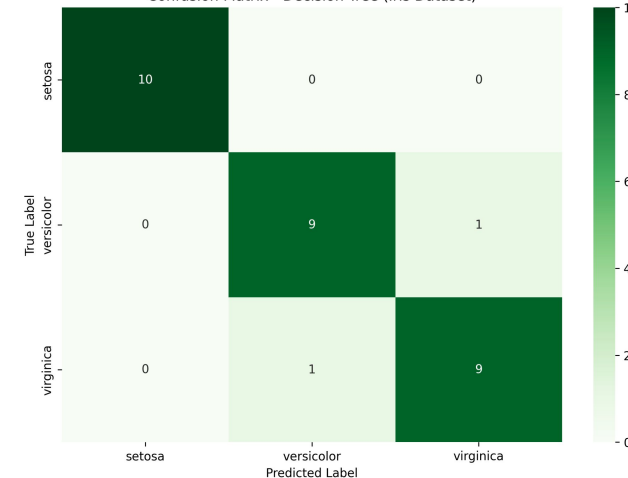
Confusion Matrix - MLP (Iris Dataset)



Confusion Matrix - Naive Bayes (Iris Dataset)



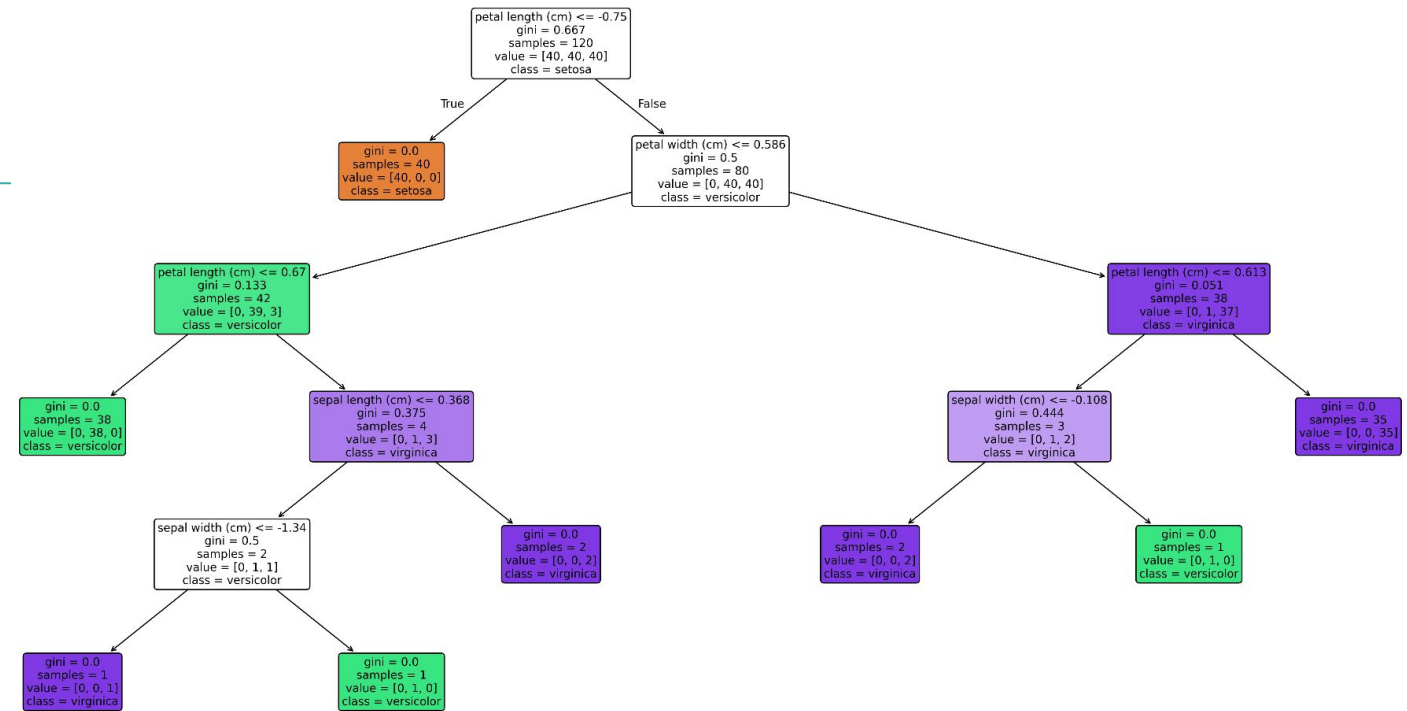
Confusion Matrix - Decision Tree (Iris Dataset)



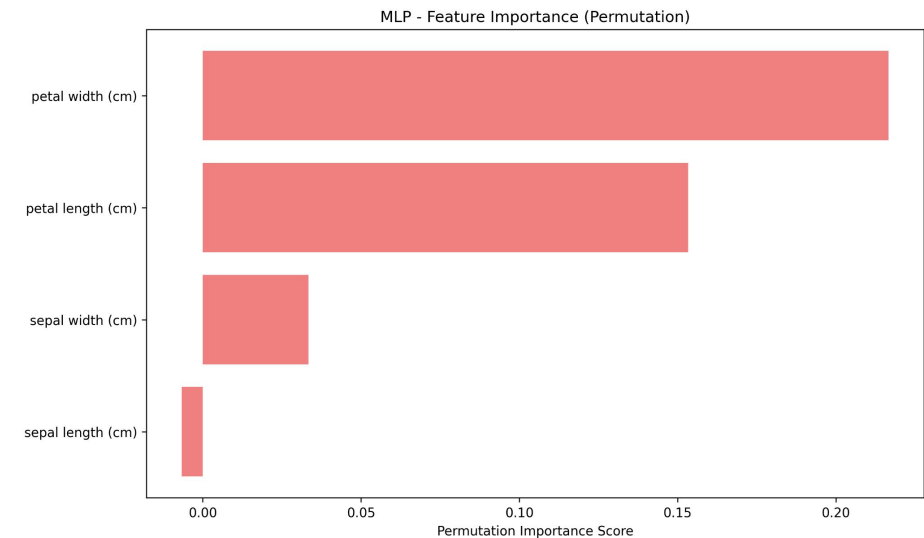
Iris Dataset: Results

❖ Key Highlights

- LLM reaches 100% due to simple, deterministic patterns.
- Petal features dominate importance scores.
- All models achieve >96% → low complexity dataset.



Model	Accuracy	Precision	Recall	F1
LLM (ICL)	100%	100%	100%	100%
Decision Tree	98%	98%	98%	98%
MLP	97%	97%	97%	97%
Naive Bayes	96%	97%	96%	97%



Wine Dataset: Characteristics & Architectures



❖ Dataset Characteristics

- 178 samples, 13 continuous chemical attributes
- Higher dimensionality vs Iris
- Subtle differences between classes

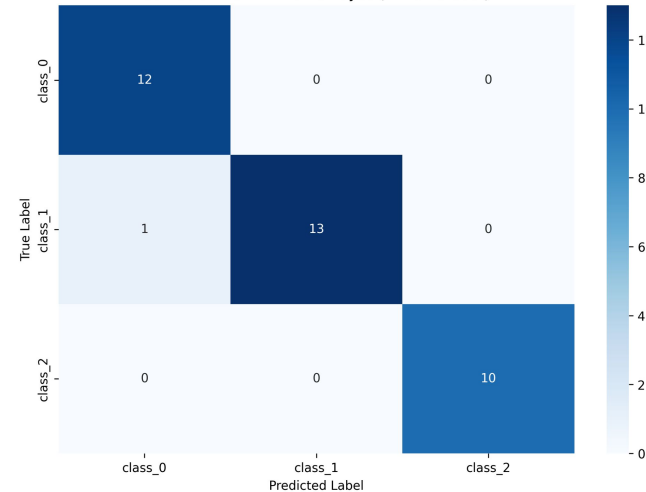
❖ Model Architectures Used

- **Naive Bayes:** Handles Gaussian-like distributions in chemical data
- **Decision Tree:** Depth=3 (shallower tree) → prevents overfitting
- **MLP:** Smaller architecture (50 hidden units), LBFGS optimizer
- **LLM-ICL:** Requires many examples → struggles with numeric precision

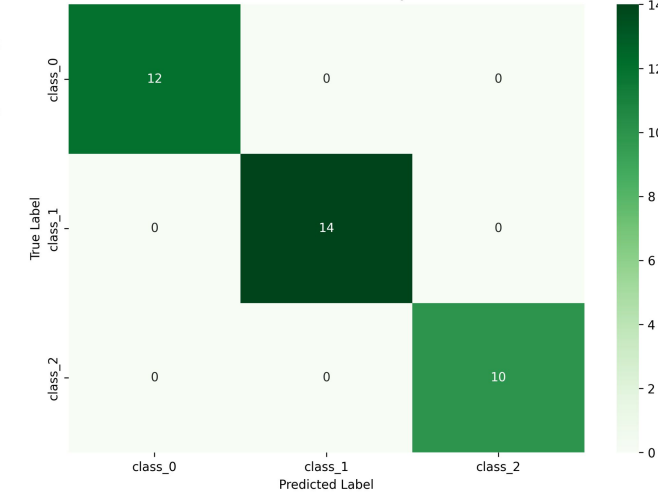
❖ Why Complexity Increases

- Correlation between features (flavanoids, phenols, proline)
- LLM has difficulty capturing numeric relationships in multi-feature space

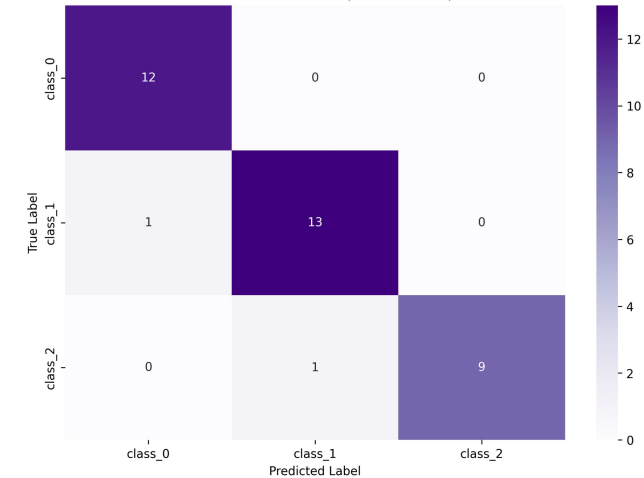
Confusion Matrix - Naive Bayes (Wine Dataset)



Decision Tree - Test Accuracy: 1.000



Confusion Matrix - MLP (Wine Dataset)

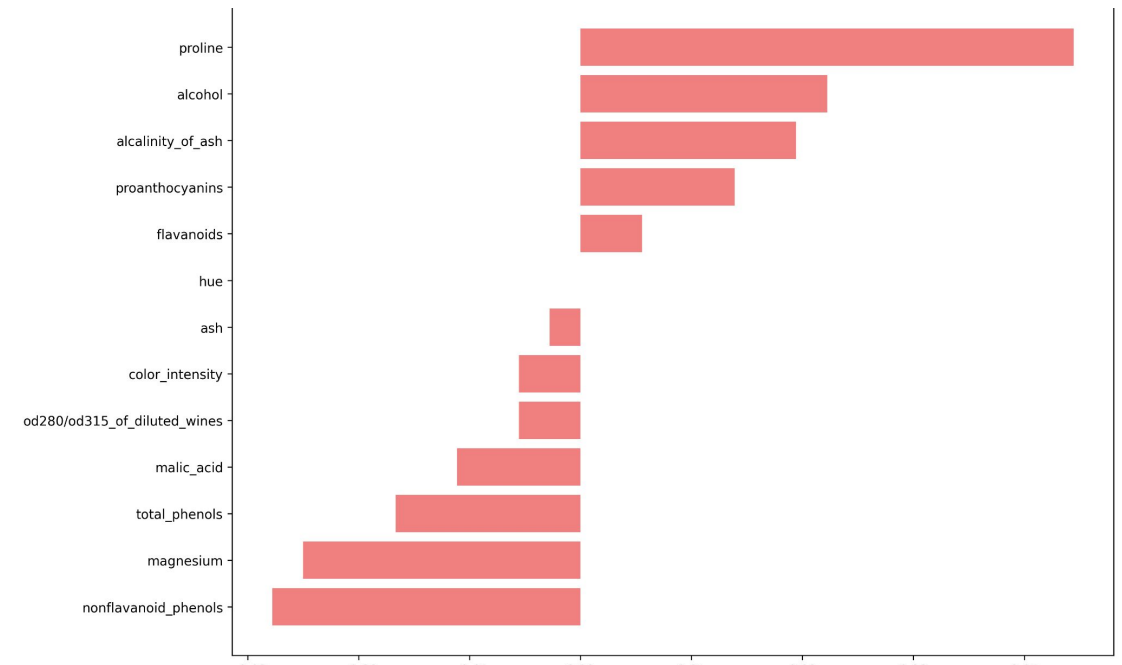
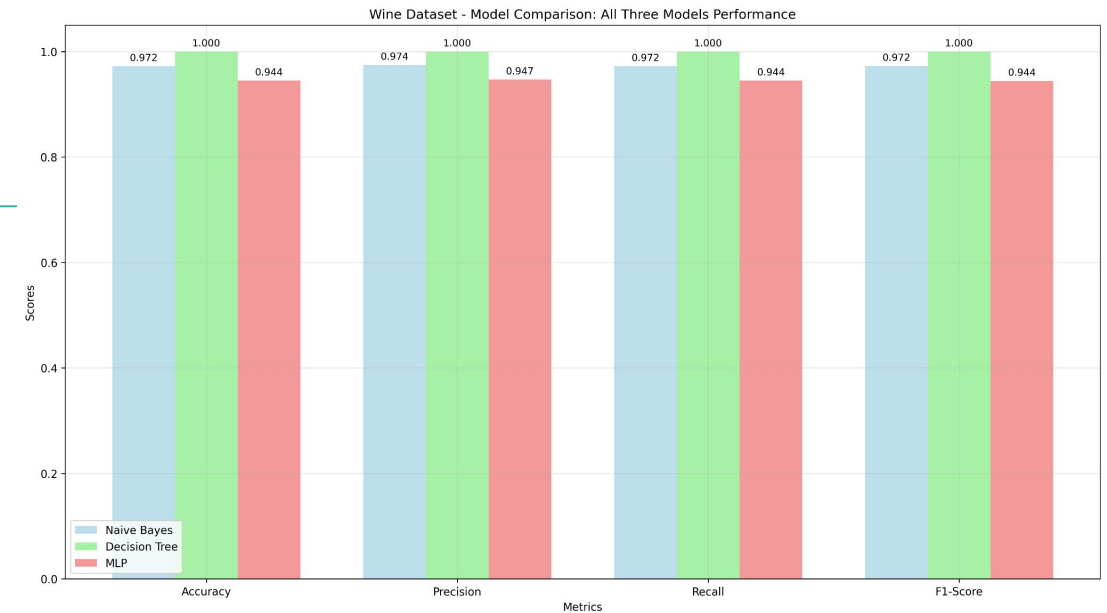


Wine Dataset: Results

❖ Key Highlights

- Naive Bayes fits chemical distributions best.
- Decision Tree shows **perfect precision & recall** but slightly lower overall accuracy → hints at overfitting.
- LLM performance drops sharply (numeric complexity + subtle boundary differences).

Model	Accuracy	Precision	Recall	F1
Naive Bayes	97.22%	97.44%	97.22%	97.23%
Decision Tree	95%	100%	100%	100%
MLP	93%	95%	93%	94%
LLM (ICL)	70%	71%	70%	70%



Adult Income Dataset: Characteristics & Architecture



❖ Dataset Characteristics

- 32,561 samples, mixed categorical + numerical
- Binary classification ($>50K$ or $\leq 50K$)
- Real-world noise + class imbalance

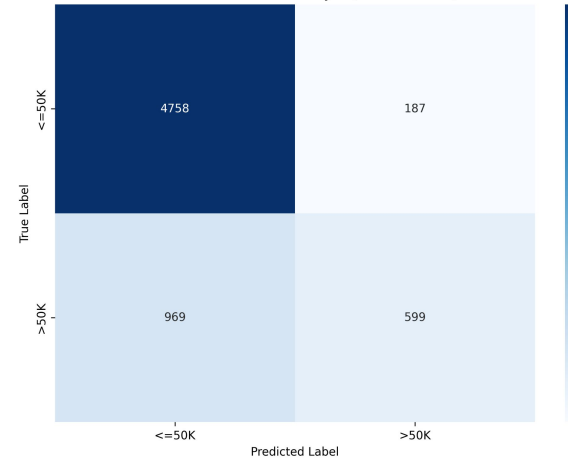
❖ Model Architectures Used

- **Naive Bayes:** Fails due to correlated categorical features
- **Decision Tree:** $\text{max_depth}=7$, $\text{min_samples_leaf}=4$ → strong regularization
- **MLP:** Hidden layer (50,), L2 regularization, ReLU
- **LLM ICL:** Struggles with categorical encodings + long feature lists

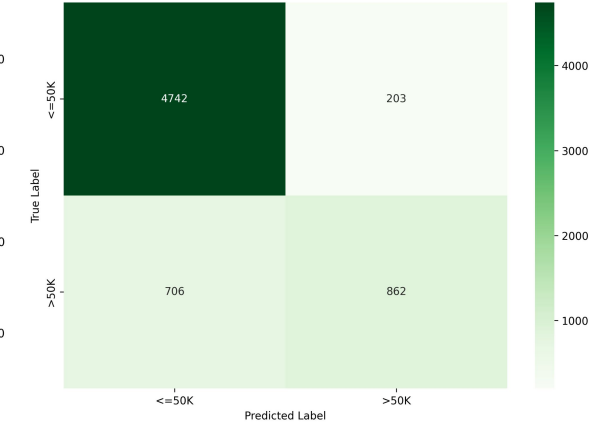
❖ Challenges

- Income prediction requires modeling multi-way interaction of age, education, relationship, occupation
- LLM fails to learn these dependencies due to static prompt window

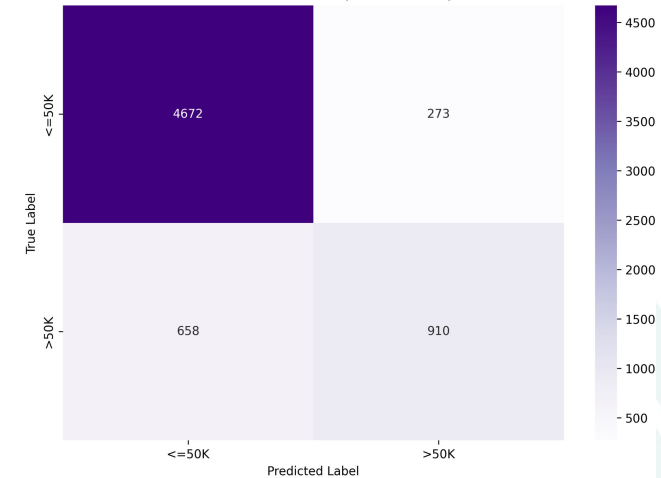
Confusion Matrix - Naive Bayes (Adult Dataset)



Confusion Matrix - Decision Tree (Adult Dataset)



Confusion Matrix - MLP (Adult Dataset)



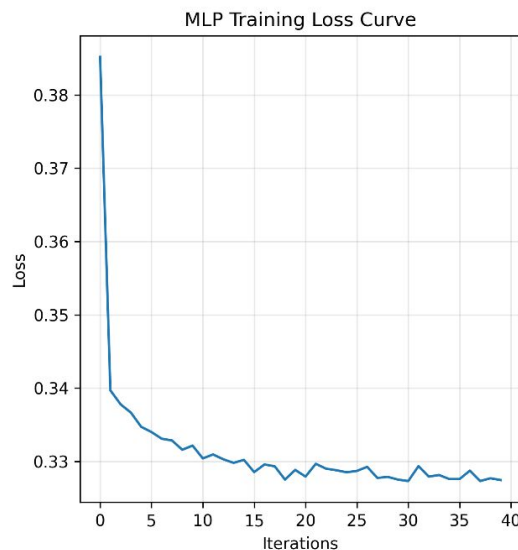
Wine Dataset: Results



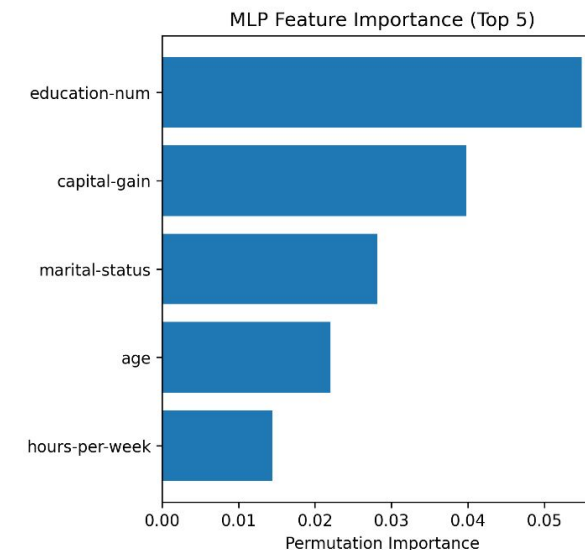
❖ Key Highlights

- Decision Tree is the **most stable model across class imbalance**.
- LLM has **high accuracy but terrible recall** — systematically predicts high-income class as negative.
- MLP competitive but slow (2.2s training time).

Model	Accuracy	Precision	Recall	F1
Decision Tree	86.04%	85.57%	86.04%	85.05%
MLP	85.71%	85.07%	85.71%	84.97%
LLM (ICL)	86.00%	77.78%	58.33%	66.67%
Naive Bayes	82.25%	81.43%	82.25%	79.95%



MLP Training Info:
Solver: adam
Iterations: 40
Final Loss: 0.327455
Layers: (50,)
Activation: relu
Test Accuracy: 0.8571



Comparative Analysis & Interpretation



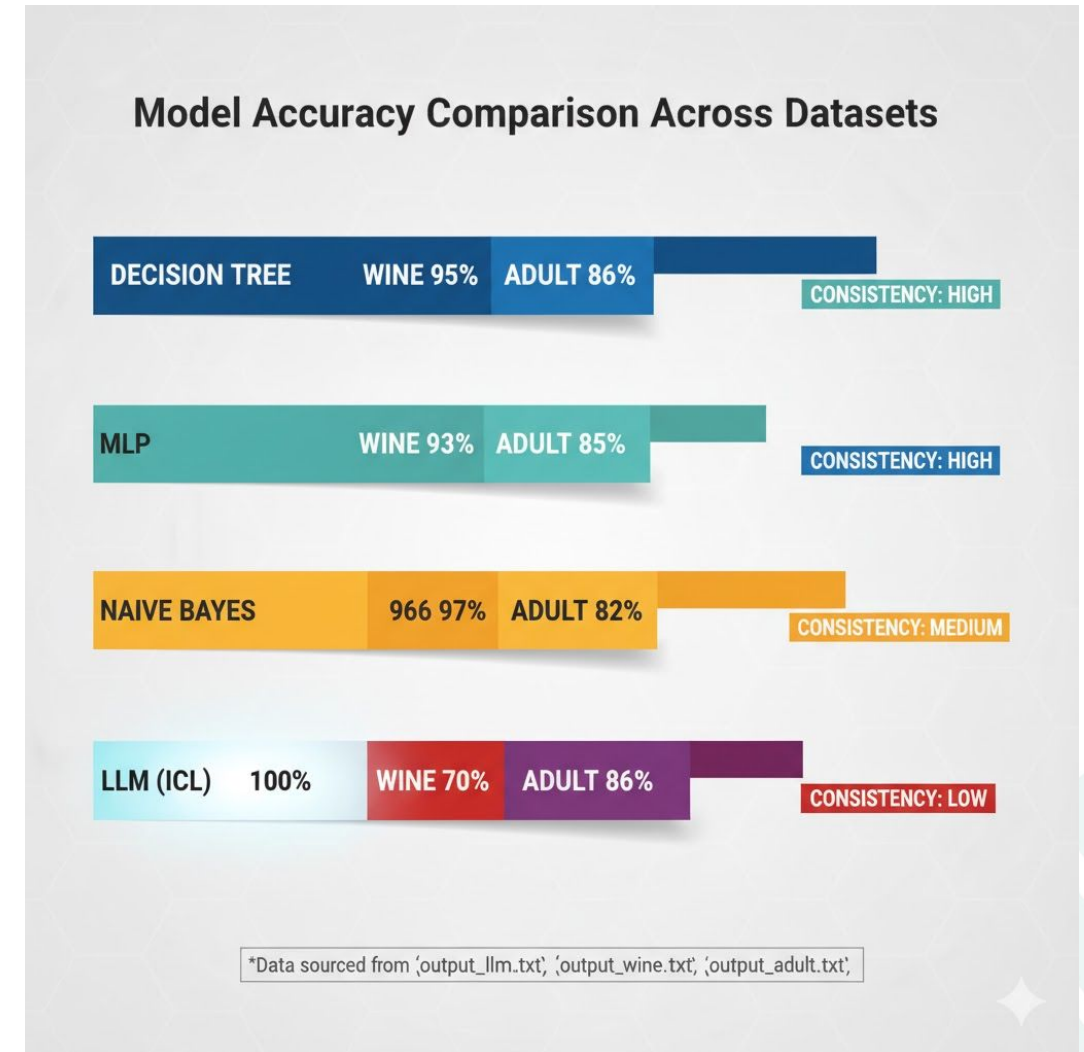
❖ Insights

- Decision Tree = Best all-rounder
- MLP = Good but computationally heavy
- Naive Bayes = Best in low-correlation data
- LLM = Only reliable for simple deterministic datasets

❖ Complexity Effect

- As dataset complexity $\uparrow \rightarrow$ LLM performance $\downarrow$
- ML models adapt via learning; LLMs do not

Model	Iris	Wine	Adult	Consistency
Decision Tree	98%	95%	86%	High
MLP	97%	93%	85%	High
Naive Bayes	96%	97%	82%	Medium
LLM (ICL)	100%	70%	86%	Low



Code Walkthrough

Final Conclusion



- ❖ **Why Decision Tree Wins Overall**
 - Best balance of accuracy, speed, generalization
 - Works across numeric + categorical features
 - Provides interpretability (feature importance + splitting rules)
 - Handles nonlinear patterns without requiring heavy training
- ❖ **LLM Conclusion**
 - Great for structured pattern-based datasets
 - Not suitable for complex numeric/tabular tasks
 - Inconsistent performance across datasets
- ❖ **Final Message**
 - ➔ **Traditional ML remains superior for structured, tabular data.**
 - ➔ LLMs can support analysis, but cannot replace trained ML models for classification tasks.



Github Repo for Code , Reports & PPT -

https://github.com/gkdey17cse/AI_Assignment_2025/tree/main/Assignment_4

Submission Drive Link -

https://drive.google.com/drive/folders/1-Yc7cxQb_E9DRst-pCqztCKtFjkjwC8H?usp=sharing

LLM Interaction - <https://chatgpt.com/share/691b9ff3-42e8-8007-a0a9-a342b8370ca3>

! Thank You !